



FACULTY OF INFORMATION TECHNOLOGY

ICS 1204: INTEGRAL CALCULUS

Date: 24th July 2018

Time: 2 Hours

Instructions

1. This examination consists of **FIVE** questions.
2. Answer **Question ONE (COMPULSORY)** and any other **TWO** questions.

Question One (30 marks)

a) Evaluate the following indefinite integrals

i. $\int \left(\frac{x^2 - 1}{x} \right)^2 dx$ (3 marks)

ii. $\int x^2 \sin x \, dx$ (4 marks)

iii. $\int \frac{5x - 10}{(x - 4)(x + 1)} \, dx$ (4 marks)

b) Evaluate the following definite integral

i. $\int_0^{\pi} \sin^3 x \cos^2 x \, dx$ (3 marks)

ii. $\int_0^1 x^3 \sqrt{x^4 + 6} \, dx$ (4 marks)

c) Find the Taylors series expansion at $x = \pi$ the function $f(x) = \sin x$ (4 marks)

c) Find the trapezoidal and Simpson's approximation $\int_0^{12} y \, dx$ when y is given in terms of x by the following table (4 marks)

x	0	3	6	9	12
y	100	97	90	78	55

- d) Find the volume V of the solid that is obtained when the region under the curve $y = \sin^2 x$ over the interval $[0, \pi]$ is revolved about the x axis. (4 marks)

QUESTION 2: (20 MARKS)

- a) i Obtain the reduction formulae for $I_n = \int \cos^n x \, dx$ (7 marks)
- ii. Hence evaluate $\int \cos^3 x \, dx$ (3 marks)
- b) Use tabular method to evaluate the integral $\int (x^2 - x) \cos x \, dx$
(Hint choose $u = x^2 - x$ and $dv = \cos x$) (5 marks)
- c) Evaluate the integral $\int \cos^5 x \, dx$ (5 marks)

Question Three (20 marks)

- a) State the order and degree of the ordinary differential equation. Is the equation homogeneous or non-homogeneous?

$$\left(\frac{d^3 y}{dx^3}\right)^4 + 5x\left(\frac{dy}{dx}\right)^2 - y = 0 \quad (3 \text{ marks})$$

- b) Solve the differential equation using separation of variables technique.

$$(1 + x^2) dy = 3y^2 dx \quad y(2) = 10 \quad (7 \text{ marks})$$

- c) Calculate the trapezoidal and Simpsons rule approximation to

$$\int_2^4 \sqrt{x^2 + 4} \, dx \quad \text{with } n = 10. \quad (10 \text{ marks})$$

Question Four (20 marks)

- a) Resolve $\frac{10x - 2x^2}{(x-1)^2(x+3)}$ into partial fractions. Hence evaluate $\int_{-1}^2 \frac{10x - 2x^2}{(x-1)^2(x+3)} \, dx$
(8 marks)

- b) Evaluate the following integrals

i. $\int x^5 \ln x \, dx$ (4 marks)

ii. $\int \frac{2x}{\sqrt{x^2+5}} dx$ (4 marks)

iii $\int \sin^4 x dx$ (4 marks)

Question Five (20 marks)

- a) Find the first four terms of the Maclaurins series expansion of $f(x) = e^x$. (4 marks)
- b) Find the volume of the solid generated by revolving the area enclosed between $y = x^2$ and $y = x^3$ is rotated about the x axis. (6 marks)
- c) Find the area enclosed by the curves $y = x^2$ and $y = x + 6$ (5 marks)
- d) Determine the length of the curve $y = 3x^{3/2} - 1$ from $x = 0$ to $x = 1$. (5 marks)